

Regression Analysis (II) Midterm Exam.

October 16, 2017

- 1.(40) Let $\hat{\boldsymbol{\theta}}$ be an estimator of $\boldsymbol{\theta} \in R^p$, and let $MSE(\hat{\boldsymbol{\theta}}) = E[(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta})^t(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta})]$. Show that $MSE(\hat{\boldsymbol{\theta}}) = tr[Cov(\hat{\boldsymbol{\theta}})] + b(\hat{\boldsymbol{\theta}})^t b(\hat{\boldsymbol{\theta}})$, where $b(\hat{\boldsymbol{\theta}})$ is the bias of $\hat{\boldsymbol{\theta}}$.
- 2.(40) Show that the Kullback-Leibler distance is always nonnegative.
- 3.(40) Assume that we want to compare heights of adults in Korea, China, and Japan. Using the dummy variables, describe the appropriate regression model.
- 4.(120) Consider a 2nd order polynomial regression model with 2 covariates.
 - (1)(40) Describe the model in matrix form.
 - (2)(40) Find the stationary point.
 - (3)(40) Explain when the saddle point occurs.
- 5.(40) Define the Box-Cox transformation model, and derive the likelihood function.
- 6.(120) Consider a simple linear regression model with no intercept $Y_i = \beta X_i + \epsilon_i$, $i = 1, \dots, n$, $\epsilon_i \sim N(0, \sigma^2)$.
 - (1)(40) Derive a test statistic for $H_0 : \beta = 0$.
 - (2)(40) Find a $100 \times (1 - \alpha)\%$ C.I. for a particular response $Y(x)$ at $X = x$.
 - (3)(40) Consider an inverse regression problem. First, find a point estimator $\hat{x}_0$ when the response y_0 is given. Also, find a $100 \times (1 - \alpha)\%$ C.I. for $\hat{x}_0$.